

论述题:

What is the shoulder complex comprised of?

The shoulder complex connects the upper arm to the thorax. It comprises four major bones: the clavicle, scapula, humerus, and sternum. It consists of three anatomical joints—the Glenohumeral (GH) joint, the Acromioclavicular (AC) joint, and the Sternoclavicular (SC) joint—as well as a functional “floating” joint known as the Scapulothoracic (ST) articulation.

What are the movements of the shoulder girdle?

The shoulder girdle allows a wide range of movements that are essential for positioning the upper limb.

1. Elevation
2. Depression
3. Protraction
4. Retraction (Adduction of the scapula)
5. Upward rotation
6. Downward rotation

What is the shoulder girdle?

The shoulder girdle, also known as the pectoral girdle, consists of the clavicles and scapulae on each side of the body. It connects the upper limbs to the axial skeleton at the sternoclavicular joints. The shoulder girdle provides attachment sites for muscles and supports a wide range of upper limb movements. It allows mobility through the acromioclavicular and glenohumeral joints while being stabilized by surrounding muscles and ligaments.

What are the three ligaments that support the acromioclavicular joint and what do they connect?

The acromioclavicular joint is supported by three main ligaments: the acromioclavicular ligament, the coracoclavicular ligament, and the coracoacromial ligament. The acromioclavicular ligament connects the acromion to the clavicle, providing horizontal stability. The coracoclavicular ligament, made of the trapezoid and conoid parts, connects the coracoid process to the clavicle, offering vertical support. The coracoacromial ligament links the coracoid process to the acromion, forming an arch that protects the shoulder joint.

What are the parts of the sternum?

The sternum consists of three main parts: the manubrium, the body, and the xiphoid process. The manubrium is the upper portion, articulating with the clavicles and the first two pairs of ribs. The body is the longest part, connecting with the costal cartilages of the 2nd to 7th ribs. The xiphoid process is the small, inferior tip of the sternum, which ossifies later in life and serves as an attachment site for abdominal muscles.

What is another name for the shoulder joint? And what are the bones and movements of this joint?

Another name for the shoulder joint is the glenohumeral joint. It is formed by the articulation of the head of the humerus and the glenoid cavity of the scapula. This ball-and-socket joint allows a wide range of movements, including flexion, extension, abduction, adduction, internal rotation, external rotation, and circumduction. It is the most mobile joint in the body, stabilized by muscles, ligaments, and the glenoid labrum.

What are the parts of the clavicle?

The clavicle consists of three main parts: the sternal end, the shaft, and the acromial end. The sternal end is medial and articulates with the manubrium of the sternum, forming the sternoclavicular joint. The shaft is the elongated middle portion, slightly curved, serving as an attachment site for muscles and ligaments. The acromial end is lateral, flattened, and articulates with the acromion of the scapula, forming the acromioclavicular joint.

What is the location of the scapula in relation to the ribs and spinal column?

The scapula is a flat, triangular bone located on the posterior thoracic wall, overlying the 2nd to 7th ribs. It lies medial to the humeral head and lateral to the vertebral column, anchored by muscles to the rib cage to support upper limb movement.

Describe factors limiting all-out aerobic performances and sites of fatigue in detail.

Diet, environment (hot/cold et al. , strength/skill , CNS function, energy production from anaerobic and aerobic sources, Central and peripheral nerve fatigue , muscle fatigue in short-term performances, cross-bridge cycling

Write a note on the physiological effects of strength training.

Strength training, also known as resistance exercise, increases muscle strength by making muscles work against a weight or force. In the short term, exercise can lead to muscle fatigue and soreness, in long term, it can lead to muscle hypertrophy and other physiological changes, making repeat exercise easier. Strength training may also help develop strong bones, manage weight, enhance quality of life

Discuss diet pyramid briefly

A visual representation (in the shape of a pyramid) of the optimal number of servings of food a person should eat daily from each basic food group, bread and cereals, several covering fruits and vegetables, and meat and poultry

Briefly describe the functions of lungs.

Breathing (Ventilation) ,Gas Exchange (Respiration),Oxygen Transport , Carbon Dioxide Removal , Regulation of Acid-Base Balance, Vocalization

What is the main parameters for evaluating the muscular strength?

Strength depends on type of muscle contraction (isometric, isotonic), muscle length, fiber types, neural factors.

Define homeostasis.

Homeostasis is the process by which living organisms maintain a stable internal environment despite external changes. It involves physiological mechanisms such as temperature regulation, pH balance, and fluid control, ensuring optimal conditions for cellular function. This balance is crucial for survival and is maintained through feedback systems involving the nervous and endocrine systems.

How Your Hormones Affect Athletic Performance

Hormones play a vital role in athletic performance by regulating energy production, muscle growth, and recovery. Key hormones like testosterone, growth hormone, and cortisol influence strength, endurance, and adaptation to training. Imbalances or fluctuations can affect performance, fatigue levels, and injury risk. Proper training, nutrition, and rest help maintain hormonal balance for peak athletic results.

名词解释: Fast-twitch muscle fiber

Type II muscle fibers (IIa & IIb/IIx) are able to produce lots of force but they also tire quicker than slow-twitch muscle fibers.

名词解释: VO2 Max

The amount of oxygen taken in and used by the body, also referred to as the rate of oxygen use. Often measured using a one-minute time frame.

名词解释: Range of motion

Range of motion (ROM) refers to the full movement potential of a joint, measured in degrees, indicating flexibility, mobility, and joint health. It can be active or passive , and is influenced by muscles, tendons, ligaments, and joint structure.

名词解释: Underweight

Human body may not be getting the nutrients to build healthy bones, skin, and hair. Underweight may relate to symptoms or signs Some people may have a genetic or a medical illness , over diet is relative to underweight